



The Beach That Ate the Footprints

A science story about tides



The Sand Keeps Track

Mia and Aarav ran along the beach, leaving footprints behind them. “Look!” Mia laughed. “It’s like the sand is keeping track of us.” Aarav pressed his foot deeper into the wet sand. Soon they were busy building a sandcastle near the shore.



The Tide Mystery

Grandfather walked over with a smile. "You've discovered the tide," he said. He explained that tides are the ocean's regular rise and fall. Sometimes the water moves farther onto the beach, and sometimes it moves away again.



A Pull From the Moon

“But why does the tide happen?” Mia asked.

Grandfather drew a circle in the sand and pointed to the sky. The Moon’s gravity gently pulls on Earth’s oceans, he said. The Sun helps a little too, but the Moon does most of the work.



The Water Comes Closer

Now the children watched the rising tide. First the water reached the edge of their castle. Then it filled the moat. Then one gentle wave knocked over a tower. “The tide is coming in right now,” Grandfather said.



The Beach Gets a Clean Page

By sunset, their footprints were gone and the sandcastle had melted back into smooth sand. “So the tide didn’t eat them,” Mia said. “It washed them away.”

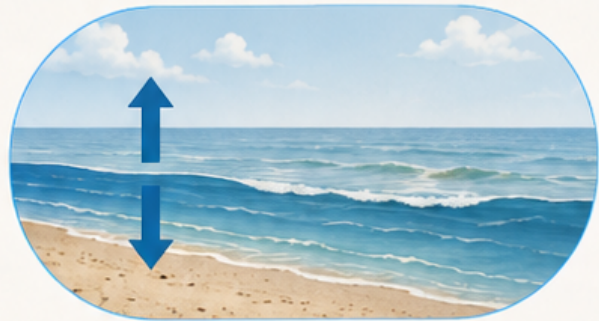
Aarav smiled. “The beach gets a clean page every day.”

Science Element



1 What are tides?

Tides are the regular rise and fall of ocean water.



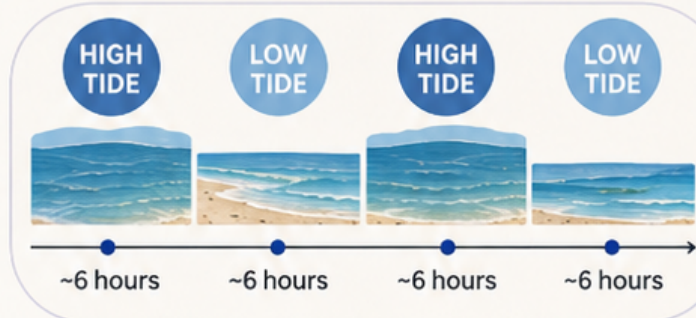
2 Why do they happen?

They happen mainly because of the Moon's gravity, and also a little because of the Sun's gravity.



3 How often?

Most beaches have about two high tides and two low tides each day.



4 Why footprints disappear

Rising water covers the sand and waves smooth it out.





The End

Mia and Aarav made one last footprint in the wet sand and waved goodbye as a gentle wave erased it. Now they knew the secret: tides keep the beach changing every day.